

Generation Interconnection System Impact Study Report

For Queue Project Q22-2032

Hughes 161 kV
August 28, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2032. The Interconnection Customer proposes a 259.9 MW battery energy storage facility interconnecting at the Hughes 161 kV substation in the New York area. The request was received on March 14, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 594 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 43 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$69,281,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Athenia / Dam Hts 138 KV is overloaded to 209.2% of its 214 MVA emergency rating for the loss of the Artesian / Wasco 138 KV. The project contributes 7.9% DFAX (20.5 MW) to this overload.

For the loss of Dam Hts / Babine Lake 138 KV line, the Nietos Substation / Colorado 138 KV loads to 183.0% of its 192 MVA rating. Project DFAX: 7.4%.

Wells River / Hughes 161 KV L: 178.6% loading (264 MVA) following the outage of the Closter / Lee Vining 161 KV; DFAX = 68.3%.

Artesian / Wasco 138 KV: 172.8% loading (228 MVA) following the outage of the Athenia / Dam Heights 138 KV; DFAX = 7.5%.

For the loss of Lee Vining / Closter 161 KV Line, the Wells River / Gastonia Prime Power Park 161 KV loads to 159.6% of its 280 MVA rating. Project DFAX: 54.1%.

For the loss of Babine Lake / Dam Hts 138 KV, the Elkhorn / Mcknna 138 KV loads to 142.0% of its 232 MVA rating. Project DFAX: 7.4%.

Cedar Grove / Multitrade Of Pittsylvania Lp 138 KV: 140.7% loading (404 MVA) following the outage of the Multitrade Of Pittsylvania Lp / Northeastern 138 KV Line; DFAX = 6.4%.

The Indian Mesa / Wasco 138 KV is overloaded to 139.8% of its 247 MVA emergency rating for the loss of the Athenia / Dam Hts 138 KV. The project contributes 7.5% DFAX (19.4 MW) to this overload.

Wellsville / Church 138 KV: 139.3% loading (236 MVA) following the outage of the W Pt Municipal / Yahara River 138 KV; DFAX = 5.2%.

Yahara River / Colorado 138 KV: 138.0% loading (449 MVA) following the outage of the Sne - Sansum N Elec / Multitrade Of Pittsylvania Lp 138 KV; DFAX = 5.2%.

For the loss of Multitrade Of Pittsylvania Lp / Northeastern 138 KV, the Cedar Grove / Veve Calzada 2002 138 KV line loads to 136.6% of its 447 MVA rating. Project DFAX: 6.4%.

Jackson Road / Church 138 KV L: 135.0% loading (222 MVA) following the outage of the Yahara Riv / W Point Municipal 138 KV; DFAX = 5.2%.

Cowan / North Bergen 161 KV line: 134.6% loading (277 MVA) following the outage of the Hills Chapel / Wallenpaupack 345 KV; DFAX = 6.9%.

The Beeline / Montville 138 KV is overloaded to 133.5% of its 278 MVA emergency rating for the loss of the Audobon 345 kV/1 kV Transformer. The project contributes 5.9% DFAX (15.3 MW) to this overload.

Jackson Rd / Passaic Valley Water Commission 138 KV: 131.5% loading (210 MVA) following the outage of the Yahara River / W Point Municipal 138 KV Line; DFAX = 5.2%.

Lane / Passaic Vly Water Commission 138 KV: 130.7% loading (231 MVA) following the outage of the Yahara Riv / Colorado 138 KV; DFAX = 7.7%.

The Belfort 138 kV/1 kV Autotransformer is overloaded to 130.7% of its 100 MVA emergency rating for the loss of the Chickasaw / Homestead Substation 161 KV. The project contributes 5.2% DFAX (13.6 MW) to this overload.

Ft Sage Substation / Montville Substation 138 KV: 130.4% loading (274 MVA) following the outage of the Audobon 345/1 kV Auto; DFAX = 5.9%.

The Mclain / Closter 161 KV is overloaded to 130.3% of its 327 MVA emergency rating for the loss of the Lee Vining / Closter 161 KV. The project contributes 31.7% DFAX (82.5 MW) to this overload.

The Nietos / Mckenna 138 KV L is overloaded to 130.1% of its 226 MVA emergency rating for the loss of the Babine Lake / Dam Hts 138 KV. The project contributes 7.4% DFAX (19.2 MW) to this overload.

The Ft Sage / Wellsille 138 KV is overloaded to 127.2% of its 275 MVA emergency rating for the loss of the W Pt Municipal / Yahara Riv 138 KV line. The project contributes 5.2% DFAX (13.5 MW) to this overload.

San elipe / Grand Island 161 KV line: 127.1% loading (259 MVA) following the outage of the Hills Chapel / Wallenpaupack 345 KV; DFAX = 6.9%.

Dam Heights / Babine Lake 138 KV: 126.1% loading (577 MVA) following the outage of the Crawfordsville Power Plant / Pine Tree Acres 138 KV Line; DFAX = 6.5%.

Hughes / Mclain 161 KV: 125.2% loading (328 MVA) following the outage of the Lee Vining / Closter 161 KV; DFAX = 31.7%.

The Multitrade Of Pittsylvania Lp / Northeastern 138 KV is overloaded to 125.0% of its 483 MVA emergency rating for the loss of the Veve Calzada 2002 / Cedar Grove 138 KV. The project contributes 6.7% DFAX (17.5 MW) to this overload.

The W Pt Municipal / Oxford 138 KV is overloaded to 124.0% of its 354 MVA emergency rating for the loss of the Wellsville / Fort Sage 138 KV. The project contributes 6.4% DFAX (16.5 MW) to this overload.

The Closter / Selkirk Cogen 161 KV Line is overloaded to 123.5% of its 568 MVA emergency rating for the loss of the Closter / Selkirk Cogen 161 KV line. The project contributes 25.8% DFAX (67.0 MW) to this overload.

Lee Vining / Loch Mirror 161 KV: 123.2% loading (397 MVA) following the outage of the Wells Riv / Hughes 161 KV; DFAX = 47.3%.

Holloman A.F.B Substation / Cowan 161 KV L: 122.0% loading (363 MVA) following the outage of the Wallenpaupack / Hills Chapel Substation 345 KV; DFAX = 6.9%.

Lee Vining / Closter 161 KV Line: 120.9% loading (425 MVA) following the outage of the Hughes / Wells Riv Substation 161 KV; DFAX = 47.3%.

Field Street / Maywood 138 KV line: 116.0% loading (173 MVA) following the outage of the Alchem / S Tacoma 138 KV; DFAX = 8.9%.

S Tacoma / Hughes ane 138 KV Line: 115.2% loading (335 MVA) following the outage of the Dam Hts / Babine Lake 138 KV; DFAX = 7.3%.

Perrine / Pacific Power 138 KV: 114.9% loading (361 MVA) following the outage of the S Tacoma / Alchem Substation 138 KV; DFAX = 5.6%.

For the loss of S Tacoma / Alchem 138 KV, the Maywood / Pacific Power Substation 138 KV loads to 112.1% of its 291 MVA rating. Project DFAX: 7.2%.

The Achem / Field Street 138 KV is overloaded to 110.7% of its 217 MVA emergency rating for the loss of the S Tacoma / Alchem 138 KV. The project contributes 8.9% DFAX (23.0 MW) to this overload.

South Tacoma / Alchem 138 KV: 108.5% loading (400 MVA) following the outage of the Babine Lake / Dam Hts 138 KV; DFAX = 7.3%.

For the loss of Crawfordsville Power Plant / Pine Tree Acres 138 KV, the Chickasaw / Homestead 161 KV loads to 106.3% of its 267 MVA rating. Project DFAX: 5.1%.

For the loss of Crawfordsville Power Plant / Pine Tree Acres 138 KV Line, the Homestead / Old Town 161 KV line loads to 105.9% of its 195 MVA rating. Project DFAX: 5.1%.

For the loss of Hughes / Wells Riv 161 KV, the Loch Mirror Substation 161 kV/138 kV Transformer Bank loads to 105.7% of its 150 MVA rating. Project DFAX: 5.7%.

Loch Mirror 161 kV/138 kV Auto T3: 105.7% loading (150 MVA) following the outage of the Wells Riv / Hughes 161 KV; DFAX = 5.7%.

The Loch Mirror 161/138 kV Autotransformer Bank 4 is overloaded to 105.7% of its 150 MVA emergency rating for the loss of the Wells River / Hughes 161 KV. The project contributes 5.7% DFAX (14.8 MW) to this overload.

San Felipe / Old Town 161 KV: 101.6% loading (247 MVA) following the outage of the Crawfordsville Power Plant / Pine Tree Acres 138 KV; DFAX = 5.1%.

The Chickasaw / Los Coches 161 KV L is overloaded to 100.7% of its 259 MVA emergency rating for the loss of the Pine Tree Acres / Crawfordsville Power Plant 138 KV. The project contributes 5.1% DFAX (13.3 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Athenia / Dam Hts 138 KV	Rebuild line with higher-capacity conductor	\$1,613,000	34% (\$555,000)
Nietos Substation / Colorado 138 KV	Rebuild line with higher-capacity conductor	\$803,000	33% (\$266,000)
Wells River / Hughes 161 KV L	Rebuild line with higher-capacity conductor	\$2,156,000	89% (\$1,910,000)
Wasco / Artesian 138 KV	Rebuild line with higher-capacity conductor	\$1,229,000	24% (\$295,000)
Wells Riv / Gastonia Prime Power Park 161 KV	Rebuild line with higher-capacity conductor	\$833,000	82% (\$682,000)
Elkhorn / Mckenna 138 KV L	Rebuild line with higher-capacity conductor	\$800,000	33% (\$266,000)
Cedar Grove / Multitrade Of Pittsylvania Lp 138 KV	Rebuild line with higher-capacity conductor	\$2,875,000	23% (\$669,000)
Indian Mesa / Wasco 138 KV	Rebuild line with higher-capacity conductor	\$1,377,000	24% (\$331,000)
Wellsville / Church 138 KV	Rebuild line with higher-capacity conductor	\$2,524,000	36% (\$920,000)
Yahara Riv / Colorado Substation 138 KV	Rebuild line with higher-capacity conductor	\$763,000	16% (\$125,000)
Cedar Grove / Veve Calzada 2002 138 KV line	Rebuild line with higher-capacity conductor	\$6,421,000	23% (\$1,495,000)
Jackson Road / Church 138 KV line	Rebuild line with higher-capacity conductor	\$2,557,000	36% (\$932,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cowan / North Bergen 161 KV line	Rebuild line with higher-capacity conductor	\$2,033,000	100% (\$2,033,000)
Beeline / Montville 138 KV	Rebuild line with higher-capacity conductor	\$3,810,000	39% (\$1,470,000)
Jackson Rd / Passaic Vly Water Commission 138 KV	Rebuild line with higher-capacity conductor	\$1,329,000	36% (\$485,000)
Lane / Passaic Vly Water Commission 138 KV	Rebuild line with higher-capacity conductor	\$2,591,000	38% (\$991,000)
Belfort 138 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$11,133,000	100% (\$11,133,000)
Ft Sage Substation / Montville Substation 138 KV	Rebuild line with higher-capacity conductor	\$1,659,000	39% (\$640,000)
Mclain / Closter 161 KV	Rebuild line with higher-capacity conductor	\$9,955,000	78% (\$7,793,000)
Nietos / Mckenna 138 KV L	Rebuild line with higher-capacity conductor	\$2,427,000	33% (\$805,000)
Ft Sage / Wellsille 138 KV	Rebuild line with higher-capacity conductor	\$4,239,000	36% (\$1,546,000)
San elipe / Grand Island 161 KV line	Rebuild line with higher-capacity conductor	\$492,000	100% (\$492,000)
Dam Heights / Babine Lake 138 KV	Rebuild line with higher-capacity conductor	\$795,000	14% (\$109,000)
Hughes / Mclain 161 KV	Rebuild line with higher-capacity conductor	\$2,561,000	78% (\$2,005,000)
Multitrade Of Pittsylvania Lp / Northeastern 138 KV	Rebuild line with higher-capacity conductor	\$5,127,000	24% (\$1,241,000)
Oxford / W Pt Municipal 138 KV L	Rebuild line with higher-capacity conductor	\$6,581,000	35% (\$2,291,000)
Closter Substation / Selkirk Cogen 161 KV Line	Rebuild line with higher-capacity conductor	\$2,452,000	67% (\$1,642,000)
Lee Vining / Loch Mirror 161 KV	Rebuild line with higher-capacity conductor	\$2,511,000	80% (\$2,019,000)
Cowan / Holloman A.F.B 161 KV	Rebuild line with higher-capacity conductor	\$3,185,000	100% (\$3,185,000)
Lee Vining / Closter 161 KV	Rebuild line with higher-capacity conductor	\$3,168,000	80% (\$2,548,000)
Field Street / Maywood 138 KV line	Rebuild line with higher-capacity conductor	\$1,731,000	21% (\$367,000)
S Tacoma / Hughes ane 138 KV Line	Rebuild line with higher-capacity conductor	\$1,127,000	16% (\$174,000)
Perrine / Pacific Power 138 KV	Rebuild line with higher-capacity conductor	\$644,000	15% (\$97,000)
Maywood / Pacific Power Substation 138 KV	Rebuild line with higher-capacity conductor	\$1,133,000	18% (\$198,000)
Field Street / Alchem 138 KV Line	Rebuild line with higher-capacity conductor	\$1,256,000	21% (\$266,000)
Alchem / S Tacoma Substation 138 KV L	Reconductor line	\$1,031,000	16% (\$159,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Chickasaw / Homestead 161 KV	Reconductor line	\$510,000	100% (\$510,000)
Homestead / Old Town 161 KV line	Reconductor line	\$980,000	100% (\$980,000)
Loch Mirror 161 kV/138 kV Transformer T2	Replace transformer with higher-rated unit	\$6,833,000	84% (\$5,736,000)
Loch Mirror 161 kV/138 kV Auto T3	Replace transformer with higher-rated unit	\$4,673,000	84% (\$3,923,000)
Loch Mirror 161/138 kV Autotransformer Bank 4	Replace transformer with higher-rated unit	\$5,775,000	84% (\$4,848,000)
San Felipe / Old Town 161 KV	Reconductor line	\$863,000	100% (\$863,000)
Chickasaw / Los Coches 161 KV Line	Reconductor line	\$286,000	100% (\$286,000)
	Total allocated to project		\$69,281,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.